

CMS Internal

pp $\sqrt{s} = 5.02$ TeV (28.0 pb $^{-1}$)

Events

J/ ψ data $6.5 < p_T < 9.0$, $|y| < 1.6$ $0.180 < \text{ctau3D} < 0.210$ mm

status/cov = 0/3

—•— Data

— Fit

- - - Signal

- - - Background

 $N_{\text{sig}} = 1.122\text{e}+04 \pm 119$ $N_{\text{bkg}} = 4852 \pm 88.9$

SR entries = 11521

 $\mu = 3.0961 \pm 0.0012$ $\sigma_{\text{G1}} = 0.0372 \pm 0.0009$ $\sigma_{\text{CB1}} = 0.02496 \pm 0.0016$

2500

2000

1500

1000

500

0

-500

-1000

-1500

-2000

-2500

-3000

-3500

-4000

-4500

-5000

-5500

-6000

-6500

-7000

-7500

-8000

-8500

-9000

-9500

-10000

-10500

-11000

-11500

-12000

-12500

-13000

-13500

-14000

-14500

-15000

-15500

-16000

-16500

-17000

-17500

-18000

-18500

-19000

-19500

-20000

-20500

-21000

-21500

-22000

-22500

-23000

-23500

-24000

-24500

-25000

-25500

-26000

-26500

-27000

-27500

-28000

-28500

-29000

-29500

-30000

-30500

-31000

-31500

-32000

-32500

-33000

-33500

-34000

-34500

-35000

-35500

-36000

-36500

-37000

-37500

-38000

-38500

-39000

-39500

-40000

-40500

-41000

-41500

-42000

-42500

-43000

-43500

-44000

-44500

-45000

-45500

-46000

-46500

-47000

-47500

-48000

-48500

-49000

-49500

-50000

-50500

-51000

-51500

-52000

-52500

-53000

-53500

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-56000

-56500

-57000

-57500

-58000

-58500

-59000

-59500

-60000

-60500

-61000

-61500

-62000

-62500

-63000

-63500

-64000

-64500

-65000

-65500

-66000

-66500

-67000

-67500

-68000

-68500

-69000

-69500

-70000

-70500

-71000

-71500

-72000

-72500

-73000

-73500

-74000

-74500

-75000

-75500

-76000

-76500

-77000

-77500

-78000

-78500

-79000

-79500

-80000

-80500

-81000

-81500

-82000

-82500

-83000

-83500

-84000

-84500

-85000

-85500

-86000

-86500

-87000

-87500

-88000

-88500

-89000

-89500

-90000

-90500

-91000

-91500

-92000

-92500

-93000

-93500

-94000

-94500

-95000

-95500

-96000

-96500

-97000

-97500

-98000

-98500

-99000

-99500

-100000

-100500

-101000

-101500

-102000

-102500

-103000

-103500

-104000

-104500

-105000

-105500

-106000

-106500

-107000

-107500

-108000

-108500

-109000

-109500

-110000

-110500

-111000

-111500

-112000

-112500

-113000

-113500

-114000

-114500

-115000

-115500

-116000

-116500

-117000

-117500

-118000

-118500

-119000

-119500

-120000

-120500

-121000

-121500

-122000

-122500

-123000

-123500

-124000

-124500

-125000

-125500

-126000

-126500

-127000

-127500

-128000

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-130000

-130500

-131000

-131500

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-132500

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-133500

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-138000

-138500

-139000

-139500

-140000

-140500

-141000

-141500

-142000

-142500

-143000

-143500

-144000

-144500

-145000

-145500

-146000

-146500

-147000

-147500

-148000

-148500

-149000

-149500

-150000

-150500

-151000

-151500

-152000

-152500

-153000